Wavelet-Based Watermarking in Latent Diffusion Models

Christopher Liang

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Abstract

As generative AI models produce increasingly realistic synthetic content, robust watermarking techniques are critical for authenticating AI-generated image. This paper presents a wavelet-based watermarking approach that embeds imperceptible signatures directly into the latent noise space of diffusion models. My method uses 2D discrete wavelet transform with Haar wavelets to decompose latent noise into four frequency bands, embedding pseudo-random watermark sequences across all 4 subbands for enhanced robustness.

I evaluated my approach using Stable Diffusion 1.5, measuring performance through AUC and TPR@1%FPR across various attacks. Results demonstrate perfect detection on clean images (AUC = 1.0) and strong robustness against Gaussian noise and JPEG compression (AUC > 0.95 for all tested compression levels). However, the method shows vulnerability to rotation attacks, with performance degrading significantly beyond 1-degree rotations.

My wavelet-based latent space watermarking offers a practical solution for authenticating diffusion-generated images against common manipulations like compression and noise. While rotation attacks present challenges, the method's integration simplicity makes it a viable contribution to AI content authentication. Future work could explore hybrid approaches and different patterns to address rotation vulnerabilities.

I. Introduction

In recent years, the development of generative AI has accelerated rapidly. Models like DALL·E and Stable Diffusion are now capable of producing images that closely mimic human-created art and photography. As the quality of these AI-generated images continues to improve, they are becoming increasingly difficult to distinguish from those made by humans. More recently, advances in video generation have heightened concerns around deepfakes—highly realistic synthetic videos that can convincingly depict people saying or doing things they never actually did. The growing accessibility and sophistication of deepfake technology pose significant risks, including the potential to spread misinformation, damage reputations, and undermine trust in

digital media. This alarming capability erodes interpersonal trust, threatening not only individuals but also societal institutions, making the detection and mitigation of deepfakes an urgent priority.

Around the world, institutions are urgently responding to this development. The European Union's AI Act mandates that AI systems clearly mark their outputs as artificially generated, while California's SB 942 requires watermarking for AI-generated content starting January 2026. At the federal level, the proposed Civilian Agency AI Watermark Act (H.R. 9042) would require government agencies to use watermarked AI systems. These regulatory developments reflect the consensus that technical solutions for content authentication are essential for digital trust.

In response to these challenges, responsible model developers are exploring watermarking as a method to safeguard their intellectual property and prevent misuse. A watermarked AI image contains embedded signals that are imperceptible to the human eye but can be reliably detected by specialized algorithms. These watermarks serve as a digital signature, allowing platforms and users to verify the origin of an image. In doing so, they offer a promising tool for fighting misinformation, reducing the spread of deepfakes, and promoting transparency in AI-generated content.

Other machine-learning based detection systems are also a possible solution to this problem. Tools such as Intel’s FakeCatcher, which analyzes blood flow from video pixels, can achieve high accuracies currently. The benefit is that no watermarking on the developer’s side is needed. The problem is that as video and image generation capabilities increase, it may be increasingly hard to distinguish between AI created content and human created content. Watermarks circumvent this issue by having the developer inject info into their model outputs forcing detection to be possible.

The main research question this project tries to answer is whether or not a simple wavelet-based latent noise watermarking model can effectively authenticate AI-generated images under common real-world image manipulations.

II. Current Research

Digital watermarking techniques have been around for a long time, but many of the robust techniques tailored towards generative AI were developed after 2022, driven by AI development and concern over content authenticity.

Diffusion models are the main image generating AI models right now. Unlike earlier approaches such as generative adversarial networks (GANs), diffusion models synthesize images through a gradual denoising process that iteratively transforms random noise into highly detailed and coherent visual outputs. Diffusion models are trained by learning to predict a process of adding some noise each diffusion step so they can reverse the process and remove noise each step. Forward Diffusion is commonly known as the process of adding noise and DDIM [1] (Denoising Diffusion Implicit Models) is the method to map from a Gaussian noise vector to an image. My

approach which will be described in the next section also relies heavily on DDIM inversion to obtain an approximation of the starting noise during watermark detection.

To review notation for forward and reverse diffusion processes and DDIM sampling, a data point x_0 is iteratively corrupted by adding Gaussian noise over T steps. At step t, which goes from 0 to T,

$$q(x_t|x_{t-1}) = N(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I)$$

where β_t is the variance at step t and q(x) is the real data distribution. This process, when unrolled, lets us find that $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon,\ \epsilon \sim N(0,1)$ where $\bar{\alpha}_t = \Pi^t_{i=0}(1-\beta_i)$ In the generative (reverse) process, the model uses a deterministic procedure where it predicts the noise and reconstructs x_{t-1}:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\epsilon_\theta(x_t)$$

where $\hat{x}_0$ is the estimate of the clean image and ϵ_θ is the predicted added noise. This process can be inverted to get from x_0 to an approximate initial noise vector x_T (which maps to an image close to x_0) based on the assumption that consecutive noise steps are approximately equal, or

$$x_{t+1} - x_t \sim x_{t-1} - x_t$$

Tree-Ring Watermarking [2] introduced the approach of embedding watermarks in the Fourier space of initial noise vectors, utilizing DDIM inversion for detection. This work demonstrated that watermarks structured as ring patterns in frequency space exhibit robustness against geometric transformations including rotations, crops, and scaling operations. It is also the first truly “invisible” watermark since the watermark was applied to the latent noise. Specifically, Tree Ring watermarking starts with embedding rings . Building upon this foundation, ZoDiac [3] addressed the vulnerability of existing methods to diffusion-based attacks by embedding watermarks directly in the trainable latent space using pre-trained stable diffusion models, achieving detection rates above 98% with false positive rates below 6.4%.

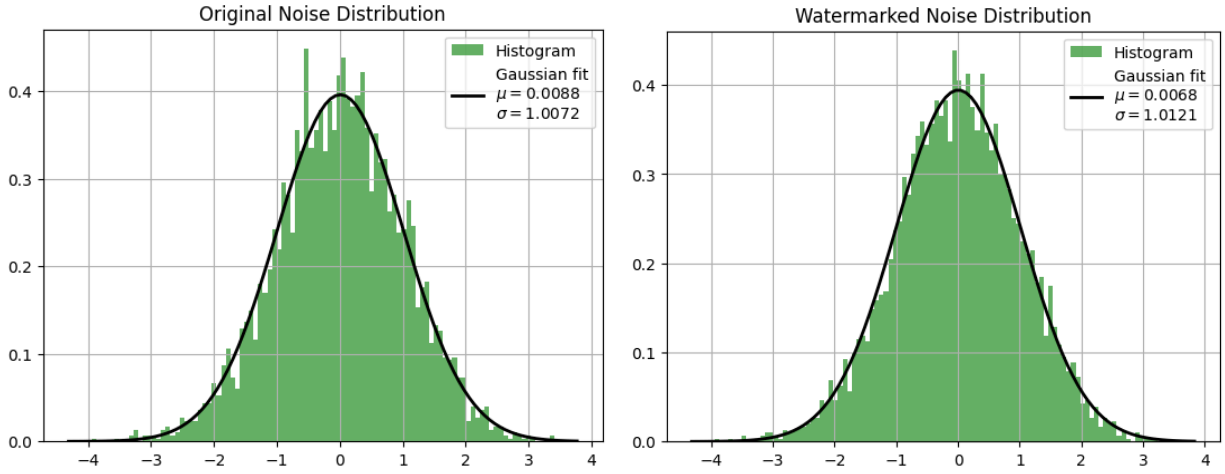
More recent approaches have explored multi-domain embedding strategies. ROBIN [4] introduced adversarial optimization techniques that embed robust watermarks in intermediate diffusion states while actively hiding them through optimized prompt guidance, demonstrating superior robustness compared to Tree-Ring methods. Similarly, FreqMark [5] proposed embedding watermarks in latent frequency space through unconstrained optimization, combining the advantages of both frequency domain and latent space approaches to achieve over 90% bit accuracy under various attack scenarios.

The problem is that once watermarking techniques improved, attacking methods also did. Regeneration attacks [6] add some noise to a generated image and regenerate it to remove pixel-level watermarks while maintaining a very similar visual look to the original image. The first

place solution to NeurIPS 2024 Invisible Watermark Removal Challenge [7] shows methods to remove the main watermark types used. A 7 px shift was employed to remove TreeRing watermarks while maintaining quality, and VAE-based watermark removal was used for StegaStamp watermarks.

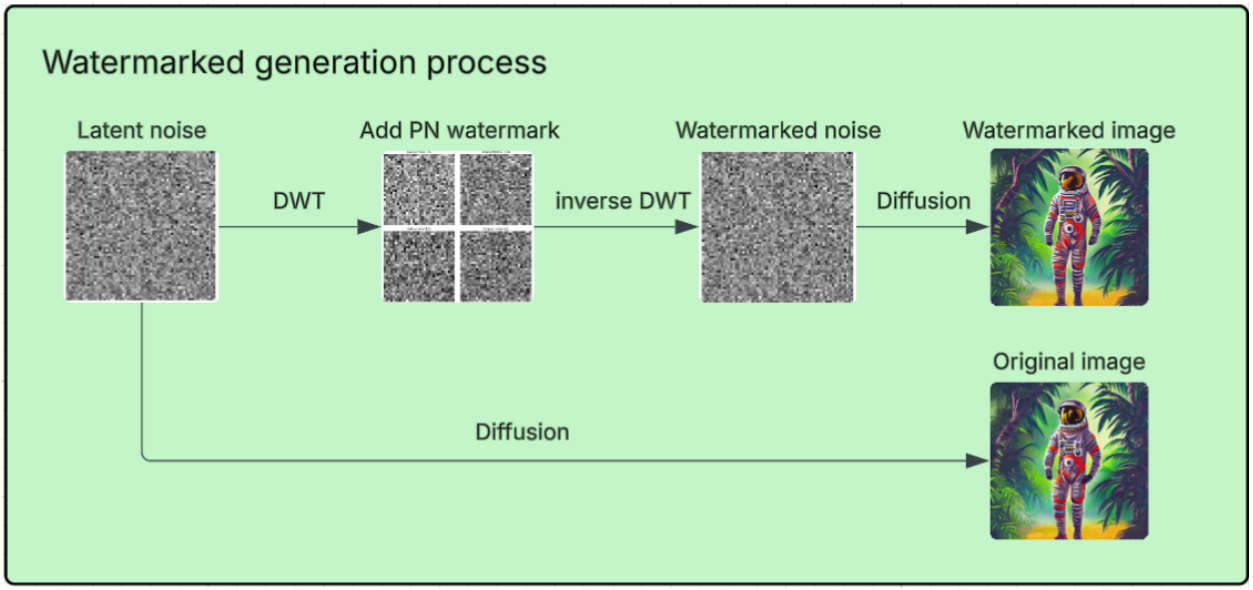
III. Approach

To achieve both an “invisible” watermark and easy integration into current diffusion models, I propose a watermarking method that embeds imperceptible signatures directly into the latent noise vector, similar to Tree-Ring [2].



Example noise distribution graphed as well as the corresponding watermarked noise, strength=0.1

The first approach I tested was the Radon transformation, which is commonly used in medical imaging and CT scans. The Radon transformation integrates the image signal along lines and has previously been proposed for watermarking[8]. However, I found that the Radon and inverse Radon transformations (not the discrete version) were too lossy for robust watermarking. The high frequency parts of the noise were very hard to preserve. Even with no watermarking, just the radon and inverse radon would create a blur effect on the noise. The blur is extremely detrimental to image quality because the AI denoising process is predicting noise based on the Gaussian distribution. As such, I decided to switch to watermarking with the 2D discrete wavelet decomposition (DWT), which didn’t have as much of an effect on the Gaussian distribution at low strengths.



Pipeline for Wavelet noise watermarking compared to a normal generation

Due to limitations in time, resources, and experience, I opted to use pseudo-random sequences (PN) of 1's and -1's generated by seeded random number generators as the watermark. Other patterns, such as concentric ring shapes, could also work for robust watermarking and could be explored in future work. This method can also be combined with others to improve robustness.

The wavelet decomposition divides the latent noise into four bands: cA, cD, cH, and cV. The cA band is the approximation band, capturing the overall structure (smooth, low frequency). The cH and cV bands capture horizontal and vertical high-frequency details, respectively, while cD captures high-frequency detail in both directions (diagonal features). Watermarks in cA should be relatively robust to JPEG compression, as JPEG typically preserves low frequencies. A 90-degree rotation swaps cH and cV and flips cD, making wavelet methods theoretically robust against "good" (orthogonal) rotations. Achieving robustness against smaller rotations (e.g., 15 degrees) is more challenging.

This project uses Haar wavelets for decomposition, primarily due to their simplicity and computational efficiency. Haar wavelets are composed of piecewise-constant basis functions, offering a fast and straightforward transform—especially useful in high-dimensional latent spaces such as those in diffusion models. Decomposing signals into approximate (low-frequency) and detail (high-frequency) bands allows watermarks to be embedded across several subbands, which may increase robustness against common image processing operations like compression and scaling.

However, Haar wavelets also have well-known limitations. Their piecewise-constant nature means that they represent smooth signal variations less effectively than more advanced wavelet families, such as Daubechies or Symlets, which have overlapping and smoother basis functions. These more sophisticated wavelets can model signal regularity more accurately and may provide improved watermark imperceptibility and resilience under certain transformations. On

the other hand, they require increased computational resources and present more delicate implementation challenges.

Given these trade-offs, Haar wavelets are used here as a practical balance of efficiency and effectiveness, but they are not necessarily the optimal choice for watermarking in latent diffusion spaces. Future work could explore alternative wavelet bases or multi-scale decompositions to further improve robustness and invisibility. Ultimately, the choice of wavelet decomposition depends on the application requirements, computational resources, and the range of attack scenarios considered.

Watermarking across all four bands helps make the watermark more imperceptible, as the modification to each band is reduced. The watermark in the cA band is particularly robust, as it tracks the general shape of the image—which is usually preserved even after severe regeneration or compression. If a ring-shaped pattern is used, greater robustness against small rotations could be achieved.

The watermark strength parameter multiplies the PN array; for example, a strength of 0.1 makes each PN element -0.1 or 0.1. Increasing the strength can boost detection rates and robustness but may degrade image quality. Conversely, a strength that is too low leads to an unstable watermark that is almost indistinguishable from noise.

IV. Theory of Change

The time and resource constraints of this project guide me towards an output-driven approach focused on evaluating a feasible watermarking method rather than pursuing an exhaustive search for the absolute best technique. Given that many major companies and organizations are actively developing watermarking solutions, contributing a diverse spectrum of methods strengthens the overall ecosystem. This diversity makes black-box attacks—where adversaries do not know the specific watermark applied—significantly more difficult.

With rising regulatory and societal pressure, exemplified by legislative initiatives such as the proposed Civilian Agency AI Watermark Act (H.R. 9042), AI developers are increasingly incentivized or required to integrate watermarking into their systems. This project's evaluation of a practical, wavelet-based watermarking approach provides critical insight into the viability and robustness of such techniques within this landscape.

As AI developers adopt watermarking methods broadly, the cumulative effect will be an AI-generated content environment where provenance and authenticity can be more reliably established. This, in turn, empowers platforms, regulators, and end users to better detect and mitigate misuse, misinformation, and unauthorized content generation, thus enhancing trust in

the digital ecosystem and eventually leading to a strong watermarking method made by combining existing methods to cover the weaknesses of each.

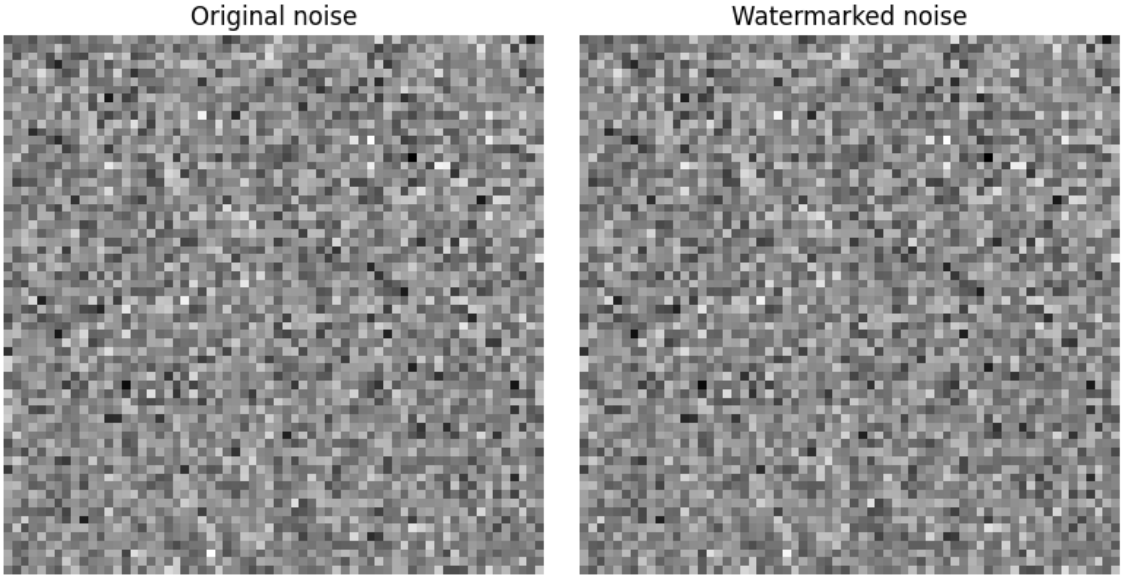
V. Methodology

The project started on a Google Colab notebook to quickly test the feasibility of this watermarking method, and was later moved on to a NVIDIA GeForce RTX 2080 Ti for faster image generation.

Google Colab is a free cloud-based platform that allows you to write and run Python code in a web browser. It's especially useful for machine learning, data analysis, and collaborative research.

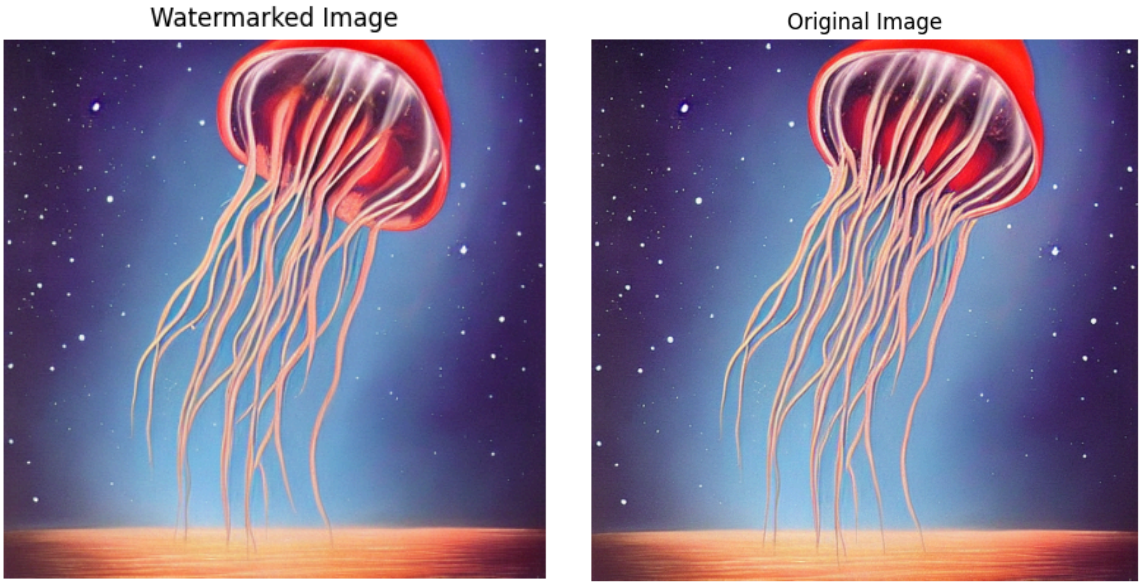
Embedding process:
We embed spread-spectrum watermarks directly into the latent noise to be used in diffusion. For each image, a pseudo-random watermark sequence is generated, multiplied by the watermark strength and added to all four bands (approximate, horizontal, vertical, and diagonal) of a wavelet decomposition of the latent noise channel before generation. Since the watermark sequence is pseudo-random, the key is saved for later detection.

A shortcoming of this method is that the detection process needs access to the key, though embedding the same sequence into each image solves this at the cost of possible removal of the watermark if the attacker has access to lots of generated images. The watermarking strength is selected to balance imperceptibility and detectability. After embedding the watermark, the DWT is reversed to get watermarked noise for use in the next step of generation.



Example of latent noise and corresponding watermarked latent noise, strength=0.1

Image generation:
Images were generated using the Stable Diffusion 1.5 model with DDIM sampling, both with and without embedded watermarks, to simulate positive (watermarked) and negative (unwatermarked) cases. Generation seeds and watermark patterns are different on each iteration to ensure generalizability. 50 generation steps were used in this experiment to balance image quality and runtime. An average time of 28 minutes was used to generate each image on Colab's free CPU.



Side by side comparison of an image generated from noise with and without a watermark, strength=0.1

Watermark Detection:
After image generation, detection is performed via DDIM inversion, reconstructing the original latent from the output image. The recovered latent undergoes wavelet decomposition, and detection is based on correlation between the recovered coefficients and the original watermark. A sample is positive if the average correlation across the 4 bands exceeds a decision threshold which can be computed after testing the model.

Model Benchmarking:
We measure detection performance using area under the ROC curve (AUC), true positive rate at fixed low false positive rates (e.g., TPR@1%FPR), and effect size (Cohen's d). A very low rate was chosen for false positive rate since a false accusation by the model could also be devastating towards artists creating work. Attacks—including JPEG compression, Gaussian noise, rotation, and scaling—are systematically applied to assess robustness. Each attack is verified quantitatively (PSNR, percent pixel change) and visually, with all metrics reported across sufficiently large datasets (100 watermarked and 100 original samples for baseline test of no attack, 10 for each strength of attack later).

VI. Results

The baseline test consisted of using an initial noise vector to generate a normal image and a watermarked image to ensure consistency. The seed of the PN used was also fixed for each set of 1 normal and 1 watermarked image. When the AUC and TPR@1%FPR were computed

Gaussian noise tests consisted of 10 positives and 10 negatives per strength test.

Baseline test	AUC = 1.000	TPR@1%FPR = 1.000
Gaussian noise 0.01	1.000	1.000
Gaussian noise 0.05	1.000	1.000
Gaussian noise 0.1	1.000	1.000
Gaussian noise 0.2	1.000	1.000

Rotation tests consisted of 10 positives and 10 negatives per rotation orientation.

Rotation degree	AUC	TPR@1%FPR
Rotation 1°	0.940	0.500
Rotation 5°	0.630	0.200
Rotation 15°	0.370	0.000
Rotation 45°	0.710	0.100

JPEG tests consisted of 100 positives and 100 negatives per compression level.

Compression level	AUC	TPR@1%FPR
JPEG 10%	0.9523	0.520
JPEG 30%	0.9909	0.860
JPEG 50%	0.9978	0.970
JPEG 70%	0.9979	0.990

The baseline test result returning AUC=1 and TPR@1%FPR=1 for 100 images means that when no attacks occur, detection is almost perfect. It is still not guaranteed since the experiment only ran for 100 original images and 100 watermarked images, while the typical amount is at least 1000 of each. The Cohen's D calculated by the program is 4.95 meaning the positive and negative classes were quite separate from each other when no attacks happened. A good baseline test lets us know that the method is working correctly, so we can proceed with attacks.

Since the AUC values for JPEG are above 0.99 for 30%, 50%, and 70%, my wavelet watermarking method seems to be relatively robust against JPEG compressions. The 0.860 for JPEG 30% implies that the watermark is still relatively discernible for highly lossy compressions. This may be in part due to the robustness of cA.

The Gaussian noise tests all returning with perfect AUC and TPR@1%FPR may be due to the insufficient batch sizes, but the technique is still most likely robust against Gaussian noise attacks since results were perfect. Further testing needs to be done for Gaussian noise attacks to validate the robustness.

The results for Rotating 1 degree were not too bad, though the low testing sizes are still likely leading to inaccurate results. 1 degree rotation still leads to high correlations since pixels are close to where they started. The subsequent rotations completely fail the detector. An AUC of 0.630 or 0.370 is extremely unreliable, and they most likely would go to 0.500 if batch sizes increased. Since rotations are a weakness of this technique, possible fixes include using different wavelet patterns which are symmetric, or combining wavelet watermarking with FFT Tree Ring watermarking.

VII. References

[1] Y. Song and S. Ermon, "Improved Techniques for Training Score-Based Generative Models," arXiv preprint arXiv:2006.09011, 2020.

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[7] F. Shamshad, T. Bakr, Y. Shaaban, N. Hussein, K. Nandakumar, and N. Lukas, “First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge,” Advances in Neural Information Processing Systems (NeurIPS), 2024.

[8] P. K. Dhar, R. Hasan, and T. Shimamura, “Color Image Watermarking Based on Radon Transform and Jordan Decomposition,” in Digital Image and Video Watermarking and Steganography. IntechOpen, Nov. 5, 2018. doi:10.5772/intechopen.80407.

VIII. Appendix

Code for baseline evaluation:
docs.google.com/document/d/1PoOofFo85wklQElkPBuOqa6loS0m0iUHonRWFpNqAcQ/edit?usp=sharing

Separate code for attacks:
docs.google.com/document/d/1u6e5b5sPQx4WvQH6KcNGsEezAWl0yin2lOmGvJT0XL0/edit?usp=sharing